

Introduction to Network Management and Administration

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*Mani Subramanian *"Network Management: Principles and practice"*, Addison-Wesley, 2000.

Introduction

Technical goal achievement

Economically positive result

► Definition:

Network management is the sum of all **activities** and **measures** to ensure the **effective** and **efficient** use of **network resources** in the interests of the company or the customer.

- Network management is a subdisipline of IT-Management
- IT-Management consists of
 - Network management
 - System management
 - Information management
 - Application management
 - Service management

Network Management Activities - Some Examples

- ▶ Detecting, reporting and eliminating technical errors
- ▶ Inventory of network components
- ▶ Identity management
- ▶ Account management
- ▶ Communication management (routing, switching, firewalls, etc.)
- ▶ Management of servers (e-mails, web server, FTPs, etc.)
- ▶ Room management so that only authorized persons have access to central network resources
- ▶ Security Management (port scanners, vulnerability scanners, virus scanners, log files, etc.),
- ▶ Backup Management (incremental/iterative data backups, virtual machine backups, configuration backups etc.)

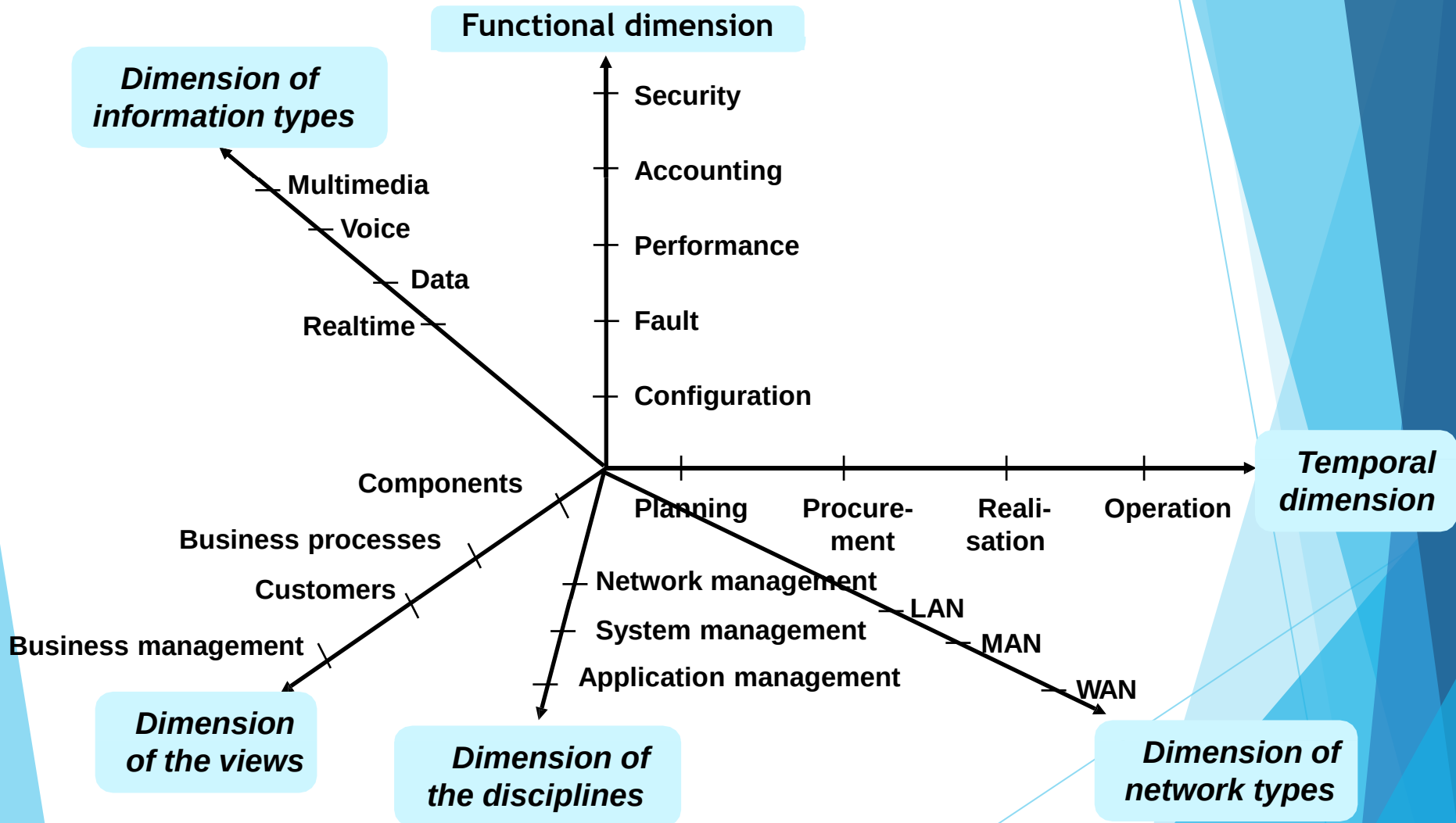
Network Management Activities - Brainstorming

- ▶ Planning
- ▶ Procurement
- ▶ Setup
- ▶ Configuration
- ▶ Controlling
- ▶ Monitoring
- ▶ Administration
- ▶ Marketing
- ▶ Testing
- ▶ Optimization
- ▶ Training

Precautions of network managements - Brainstorming

- ▶ Operational organization / procedure
- ▶ Network management tools and architectures
- ▶ Disaster planning
- ▶ User/Troubleshooting Hotline
- ▶ Backup-Service

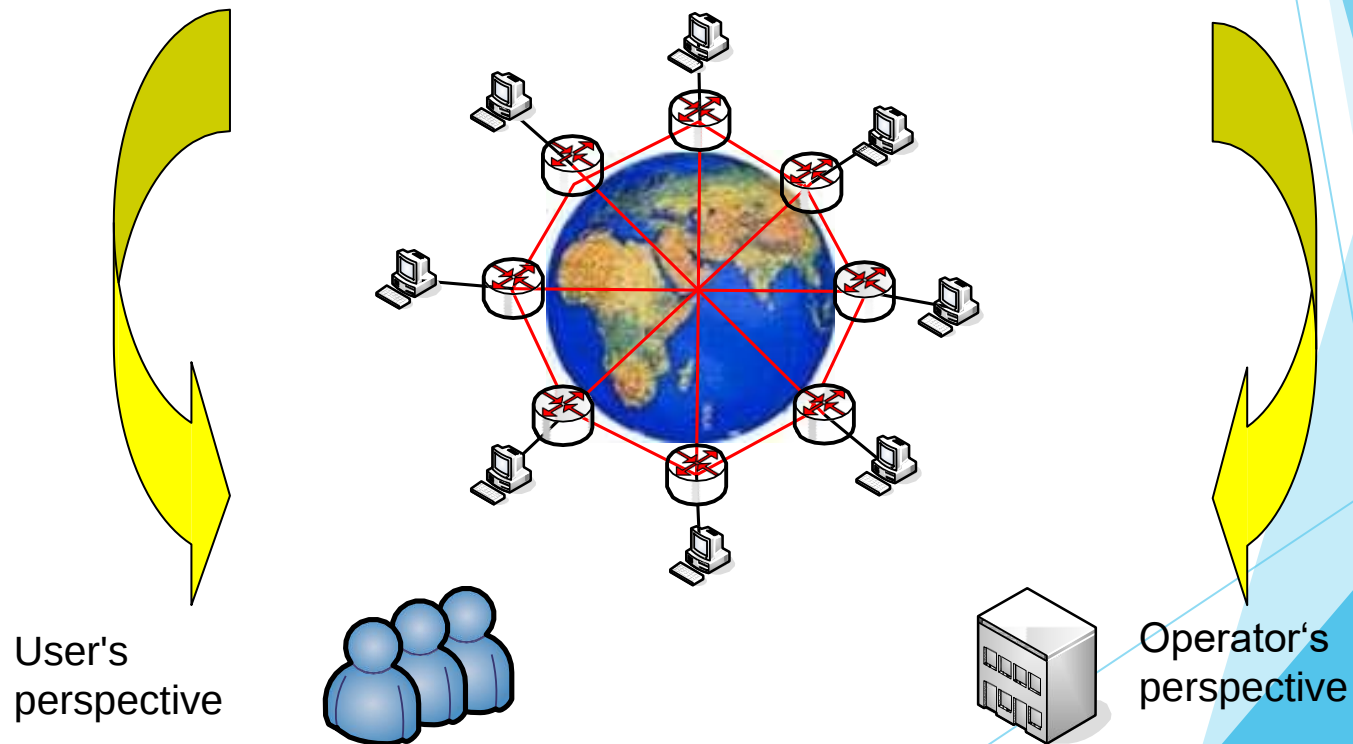
Dimensions of the management of networked systems



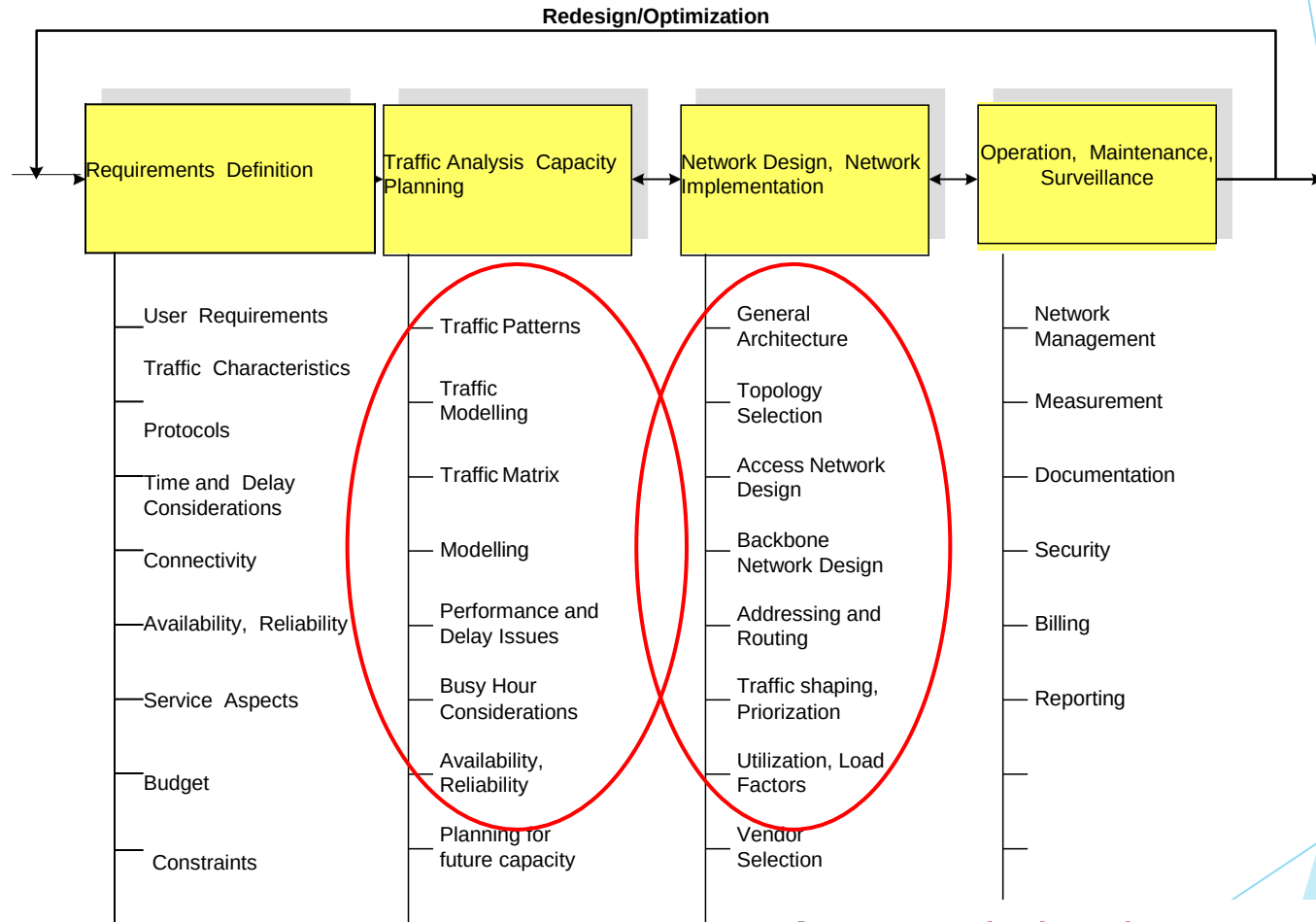
Network development and network planning

Objectives with regard to the use / provision of a communication network

- ▶ Network Monitoring
- ▶ Network Operation
- ▶ Network Development



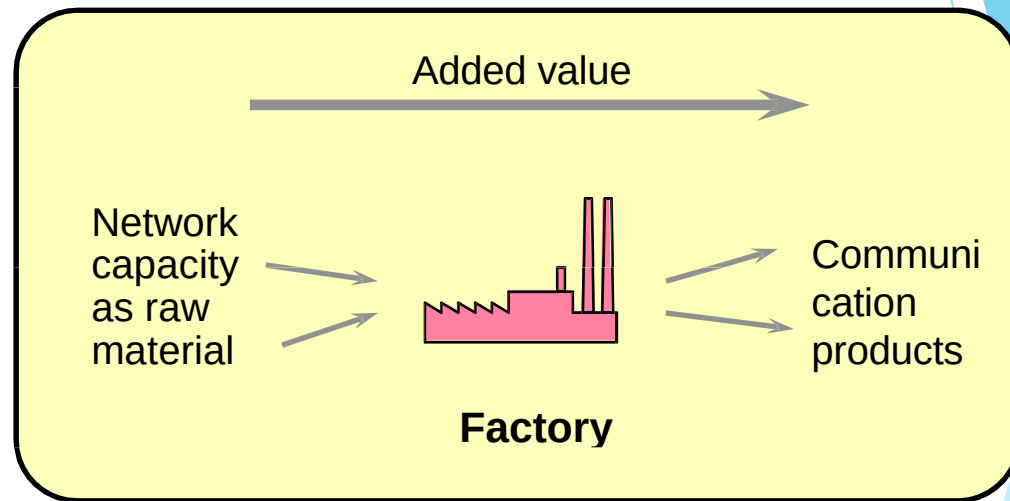
Typical steps of a network planning process



→ **Network development** requires structured, cyclical network planning, if necessary separately at different network levels, then for the entire network.

Network in the value chain

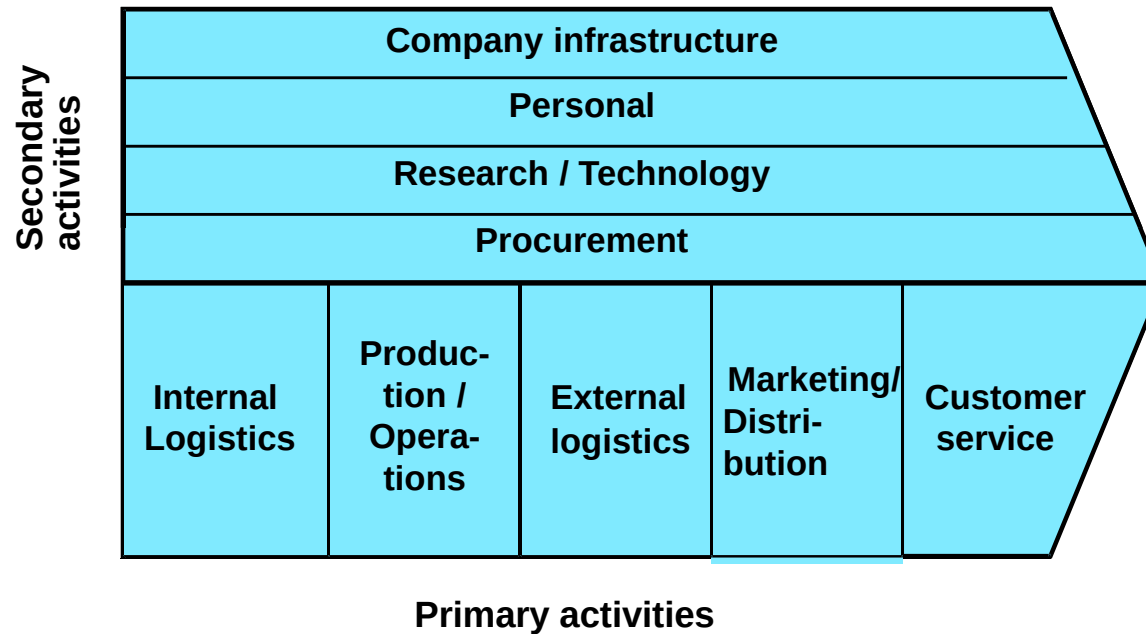
- ▶ Basic idea: The "net" factory



- Technical view on network management
- Economic point of view:
 - Network in the value chain
 - Network and IT infrastructure to support business processes

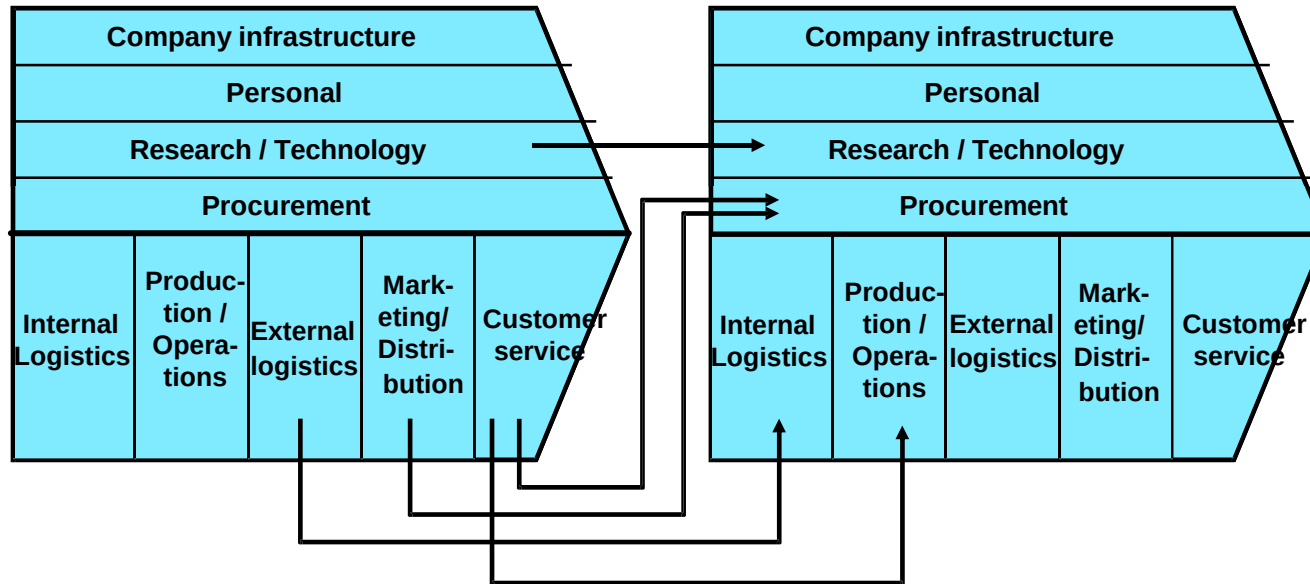
Integration of inter-company business relationships

Generic value chain according to Michael E. Porter



Integration between business relationships

Linkages between value chains according to Michael E. Porter



Business Processes

- ▶ A business process is a set of related tasks or activities performed to produce a product or service.
- ▶ Business processes exist in any type of company, independent of size or industry.
- ▶ Internal and cross-company processes are the nerve tracts of any given company. If they are clearly defined and also optimized on a regular basis, they will drive competitiveness in the long run.
- ▶ A business process describes:
 - ▶ which activities are performed in the course of a process
 - ▶ which organizational units participate in process execution
 - ▶ what input and output data are used
 - ▶ what IT systems are involved, and
 - ▶ which events and risks occur during process execution

Business process modeling

- ▶ Business process modeling is the graphic representation of a company's business processes.
- ▶ Process modeling aims at graphically describing the process flow and providing information on the company so that even complex processes remain transparent.
- ▶ The most important method of modeling a business process is the event-driven process chain (EPC).

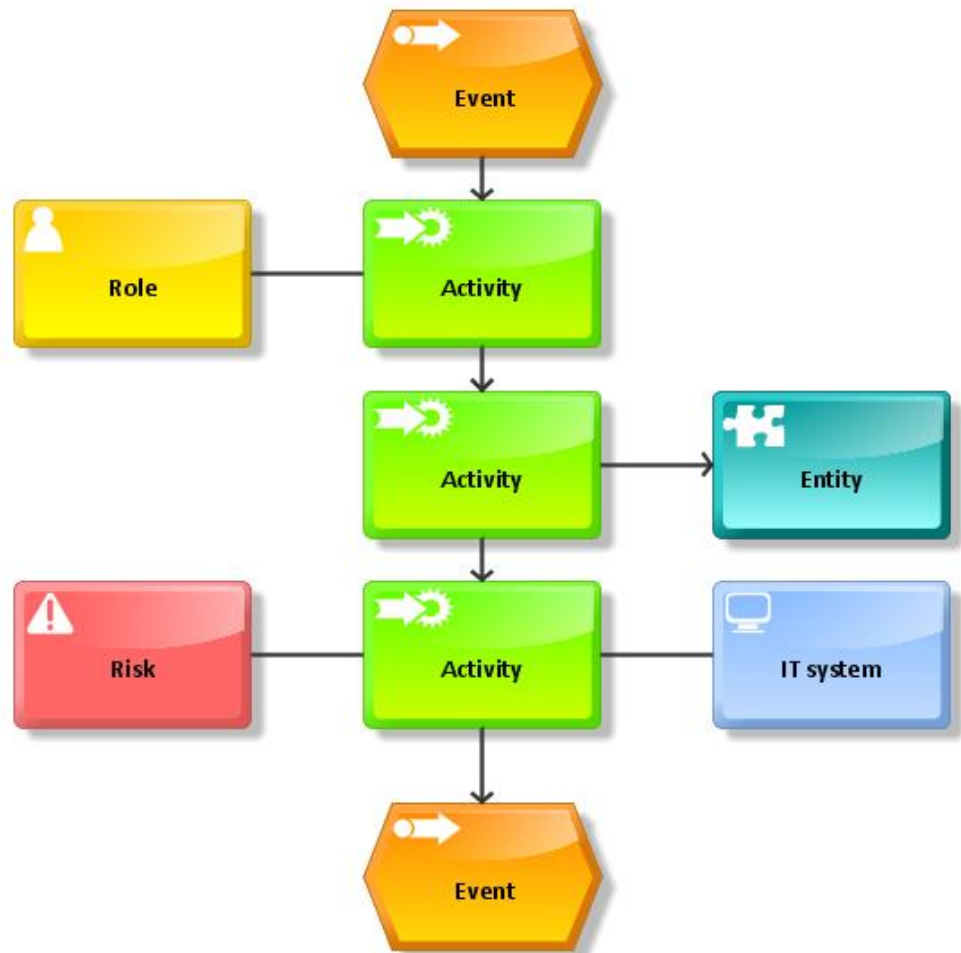


Figure. Example of a business process

Thank you for your attention!